

Treynor Ratio: What It Is, What It Shows, Formula To Calculate It

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What Is the Treynor Ratio?

The Treynor ratio, also known as the reward-to-volatility ratio, is a performance metric for determining how much excess return was generated for each unit of risk taken on by a portfolio.

Excess return in this sense refers to the return earned above the return that could have been earned in a risk-free investment. Although there is no true risk-free investment, treasury bills are often used to represent the risk-free return in the Treynor ratio.

Risk in the Treynor ratio refers to systematic risk as measured by a portfolio's beta. Beta measures the tendency of a portfolio's return to change in response to changes in return for the overall market.

Key Takeaways

- The Treynor ratio is a risk/return measure that allows investors to adjust a portfolio's returns for systematic risk.
- A higher Treynor ratio result means a portfolio is a more suitable investment.
- The Treynor ratio is similar to the Sharpe ratio, although the Sharpe ratio uses a portfolio's standard deviation to adjust the portfolio returns.

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The Treynor ratio was developed by Jack Treynor, an American economist who was one of the inventors of the Capital Asset Pricing Model (CAPM).

Understanding the Treynor Ratio

The Formula for the Treynor Ratio is:

Treynor Ratio = $\frac{r_p - r_f}{\beta_p}$ where: r_p = Portfolio return, r_f = Risk-free rate, β_p = Beta of the portfolio

What Does the Treynor Ratio Reveal?

In essence, the Treynor ratio is a risk-adjusted measurement of return based on systematic risk. It indicates how much return an investment, such as a portfolio of stocks, a mutual fund, or exchange-traded fund, earned for the amount of risk the investment assumed.

If a portfolio has a negative beta, however, the ratio result is not meaningful. A higher ratio result is more desirable and means that a given portfolio is likely a more suitable investment. Since the Treynor ratio is based on historical data, however, it's important to note this does not necessarily indicate future performance, and one ratio should not be the only factor relied upon for investing decisions.

How the Treynor Ratio Works

Ultimately, the Treynor ratio attempts to measure how successful an investment is in providing compensation to investors for taking on investment risk. The Treynor ratio is reliant upon a portfolio's beta—that is, the sensitivity of the portfolio's returns to movements in the market—to judge risk.

The premise behind this ratio is that investors must be compensated for the risk inherent to the portfolio, because diversification will not remove it.

Difference Between the Treynor Ratio and Sharpe Ratio

The Treynor ratio shares similarities with the Sharpe ratio, and both measure the risk and return of a portfolio.

The difference between the two metrics is that the Treynor ratio utilizes a portfolio beta, or systematic risk, to measure volatility instead of adjusting portfolio returns using the portfolio's standard deviation as done with the Sharpe ratio.

Limitations of the Treynor Ratio

A main weakness of the Treynor ratio is its backward-looking nature. Investments are likely to perform and behave differently in the future than they did in the past. The accuracy of the Treynor ratio is highly dependent on the use of appropriate benchmarks to measure beta.

For example, if the Treynor ratio is used to measure the risk-adjusted return of a domestic large-cap mutual fund, it would be inappropriate to measure the fund's beta relative to the Russell 2000 Small Stock index.

The Sharpe ratio is used to help investors understand the return of an investment compared to its risk.